ANDRÉ ROGÉRIO DA SILVA FILHO

Software Development Consultant | Full-Stack • Applied AI • Cloud Architecture

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PROFESSIONAL SUMMARY

Software Development Consultant specializing in full-stack architecture, autonomous AI agent systems, and cloud infrastructure, grounded in a solid technical foundation originating in enterprise IT operations and infrastructure. Acts as a technical reference in the design and delivery of software solutions using TypeScript, Python, React, Next.js, Node.js, PostgreSQL, and Google Cloud. At Grupo CBI, progressed from Full-Stack Developer to Software Development Consultant, spearheading internal platform architecture, high-availability database administration (PostgreSQL/Cloud SQL) — directly responsible for safeguarding sensitive citizen data across metropolitan public concessions under strict compliance with LGPD — and corporate AI assistant engineering (Google Gemini, MCP servers, and RAG). The transition from infrastructure to software engineering enables designing resilient, secure, and production-oriented systems.

TECHNICAL SKILLS

Languages & Frameworks: Python, TypeScript, JavaScript, Node.js, React 19/18, Next.js 15, Express, Fastify, PyTest, Vitest, Three.js, React Three Fiber, WebGL/GLSL Shaders.

Applied AI & Agents: Autonomous Agents (LangGraph, MCP Servers), LLMs (Google Gemini API), Vision-Language Models (Qwen2-VL via vLLM on AMD ROCm), PyTorch, Advanced RAG (Hybrid Search & Reranking), NetworkX Relational Graphs, Structured Outputs (JSON Schema), Zod.

Cloud, DevOps & Data: Google Cloud Platform (Cloud Run, Cloud SQL, Secret Manager), Docker, CI/CD Pipelines, PostgreSQL (Row-Level Security), Redis, Supabase, Prisma ORM, Git.

Enterprise Security & Governance: LGPD/GDPR Compliance, AES-256-GCM Encryption, IDOR Prevention, Microsoft 365, Entra ID, Microsoft Graph API, Microsoft Intune, Certified AppSec Practitioner (CAP).

PROFESSIONAL EXPERIENCE

CBI (Companhia Brasileira de Infraestrutura)

06/2026 – Present

Software Development Consultant | São Paulo, SP, Brazil (Remote) (10/2026 – Present)

- Promoted to Software Development Consultant, serving as technical reference in the design and architecture of internal platforms and AI systems for mission-critical operations across the group.
- Architects and manages cloud infrastructure on Google Cloud (Cloud Run, Cloud SQL, Secret Manager) and relational databases, directly responsible for safeguarding sensitive citizen data across metropolitan public concessions under strict LGPD compliance.
- Leads technical architecture reviews and alignments with technology and cloud vendors (including Google Cloud and Microsoft), ensuring access governance and environment resilience.
- Conceived, architected, and deployed **"Isa"**, an enterprise AI assistant integrated with Google Gemini, autonomous tool execution via MCP servers, contextual retrieval (RAG), and deterministic validation with Zod and JSON Schema.
- Conducts internal technical enablement sessions and establishes governance guidelines for safe, highly productive adoption of generative AI across engineering and business teams.

Full-Stack Developer | São Paulo, SP, Brazil (Remote) (06/2026 – 10/2026)

- Developed full-stack web applications and internal enterprise tools using TypeScript, React 19, Next.js, Node.js, and PostgreSQL.
- Implemented enterprise API integrations, webhooks, and compliance monitoring pipelines integrating telemetry from Microsoft Graph API and Microsoft Intune.
- Engineered automation routines and scripts in Python and TypeScript to streamline operations and decrease support ticket volume, accelerating promotion to Software Development Consultant.

Farmamed Drogarias

10/2025 – 03/2026

IT Infrastructure Analyst | Sorocaba, SP, Brazil

- Administered corporate IT infrastructure, identity governance, access provisioning, and endpoint compliance via Active Directory, Entra ID, and Microsoft Intune across a branch retail network.
- Developed Python automation scripts for operational data consolidation, log analytics, and proactive multi-branch diagnostics.
- Monitored critical business services and resolved high-priority technical incidents under rigorous operational SLAs.

Wobben Windpower

11/2023 – 04/2025

IT Systems & Infrastructure Intern | Sorocaba, SP, Brazil

- Selected for the global **Microsoft Copilot Flight Crew** program, evaluating generative AI features and corporate adoption workflows.
- Developed internal automation scripts and integration routines for SAP ERP, Active Directory, and Microsoft 365 in a large-scale industrial manufacturing environment.
- Delivered tier-2 specialized technical support for enterprise incidents via ServiceNow, rigorously meeting established SLAs.

SELECTED PROJECTS

Onboarding OS | Multi-Tenant Lifecycle Platform GitHub

React 19, TypeScript, Node.js, Express, PostgreSQL, Supabase, Redis, Zod, AES-256-GCM, DocuSeal

- Multi-tenant SaaS platform automating employee onboarding workflows, connecting HR, IT, and operational leaders across Grupo CBI subsidiaries.
- Implemented strict tenant isolation via PostgreSQL Row-Level Security (RLS) and transparent AES-256-GCM encryption at rest for sensitive data.
- Structured automated daily background workers for LGPD lifecycle compliance (automatic anonymization upon legal retention expiration) and state-machine validation.

Cribum (AuditRx) | Multimodal Medical Audit & Fraud Graph GitHub

AMD Developer Hackathon (Health Track) • Python, FastAPI, React 19, PyTorch, AMD ROCm, vLLM, Qwen2-VL, NetworkX, TypeScript

- Local, privacy-first multi-agent system for medical prescription and claim audits built for the AMD Developer Hackathon.
- Integrated open Vision-Language Models (Qwen2-VL via vLLM on AMD ROCm) with uncertainty-aware multi-hypothesis extraction and LASA pharmacological safety databases.
- Engineered relational multigraph anomaly detection with NetworkX to uncover suspicious billing collusion patterns and geographic billing conflicts.

Memory Lens | Cognitive Computing Platform & 3D Interface GitHub

TypeScript, React 19, Three.js, React Three Fiber, WebGL/GLSL, Express, Gemini API, Vitest

- Cognitive computing platform modeling human retention, associative recall, and structured forgetting (Ebbinghaus curve), reducing token overhead by up to 80%.
- Developed 3D neural visualization ("Chief Organism") in React Three Fiber with custom GLSL shaders synchronized to cognitive state telemetry.
- Formal academic research in semantic document organization under Prof. Dr. Fabrício Londero (UniFacens), targeted for the ACDSA international conference.

FundiVR | Industrial Digital Twin & VR Simulator GitHub

React 18, TypeScript, Fastify, PostgreSQL, Prisma ORM, WebSockets, Docker, Vitest

- Industrial digital twin simulator for training aluminum smelting furnace operators in 700°C+ environments, governed by operational safety state machines.
- Implemented real-time bidirectional telemetry via WebSockets with an automated scoring engine evaluating standard operating procedure (SOP) compliance.
- Built decoupled zero-dependency demo mode enabling standalone execution and demonstration without an active database connection.

TECHNOLOGY RESIDENCY & EDUCATION

Resident Researcher in ICT (Digital Twins & 5G) 10/2026 – Present

UniFacens • Softex • Ministry of Science, Technology and Innovation (MCTI) | ICT Residency 51

- Selected among the **top 20 out of 300 participants** from the ICT Specialization (360h) to join the Technological Residency (6-month track), working with industrial digital twins, real-time telemetry, and private 5G networks.
- Conception and development of industrial prototype FundiVR, defended and approved before an academic and industrial evaluation board.

Higher Technology Degree in Systems Analysis and Development (TADS) Graduation: 2026

Centro Universitário SENAC, Brazil

CERTIFICATIONS & LANGUAGES

Certifications: Google Cloud Generative AI Leader • Certified AppSec Practitioner (CAP) • CCNA 1: Introduction to Networks (Cisco) • Network Security & Endpoint Security (Cisco).

Languages: English (C1+ / Fluent — 2 years of immersion at Happy & Co. and practical experience in teaching and technical mentoring) • Spanish (Intermediate / Technical Communication) • Portuguese (Native).

Teaching & Mentoring: Volunteer English Instructor at Clube Poliglota Sorocaba, delivering conversation sessions and technical mentoring for international students.